A/B testing: question and one sample

Understanding Statistical Inference and Uncertainty through A/B Testing

We compare traditional teaching (A) with a new teaching method (B). All scores are synthetic. This is a statistical demonstration, not a real teaching experiment and not evidence that a teaching method works.

Plan before sampling. Estimand: population mean B minus mean A. H0: difference=0; H1: difference is nonzero. Two-sided alpha=0.05; 95% confidence intervals; minimum meaningful positive improvement=5 points. Population means are 70 and 75, both SDs are 15, and n=100 per group. The initial seed is 42; later seeds are 43, 44 and 45.

Assumptions. Independent observations within and between groups, representative samples and finite variances. The simulated populations are normal. Welch permits unequal variances. Real causal claims would also require an appropriate randomized design.

Observed sample. A: mean=69.246, SD=11.650; B: mean=74.841, SD=14.693. The mean difference is 5.595 points. Population values were chosen settings; sample statistics are realized estimates.

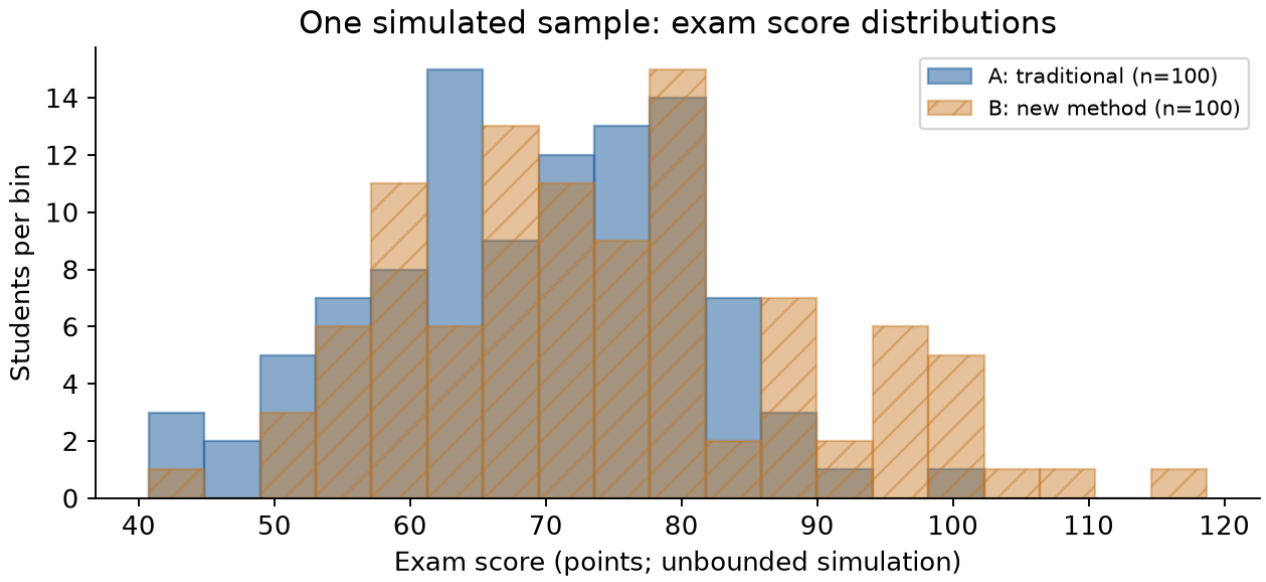


Figure 1. Both sample distributions overlap substantially. Overlap of individual scores does not itself decide whether the population means differ. Normal draws are unbounded and can exceed the usual exam scale.

Sampling variation and Welch inference

Method. With vA=sA squared/nA and vB=sB squared/nB:
SE=sqrt(vA+vB); t=(mean B - mean A)/SE.
df=(vA+vB) squared / [vA squared/(nA-1)+vB squared/(nB-1)].
95% CI = observed difference +/- t(0.975, df) x SE. Manual t, p, df and CI agree with SciPy.

Result. Difference=5.595; SE=1.875; df=188.219; p=0.003226; 95% CI [1.896, 9.293]. We reject the zero-difference null. The interval extends below 5, so an improvement of at least 5 points is not established. A p-value is not the probability that H0 is true.

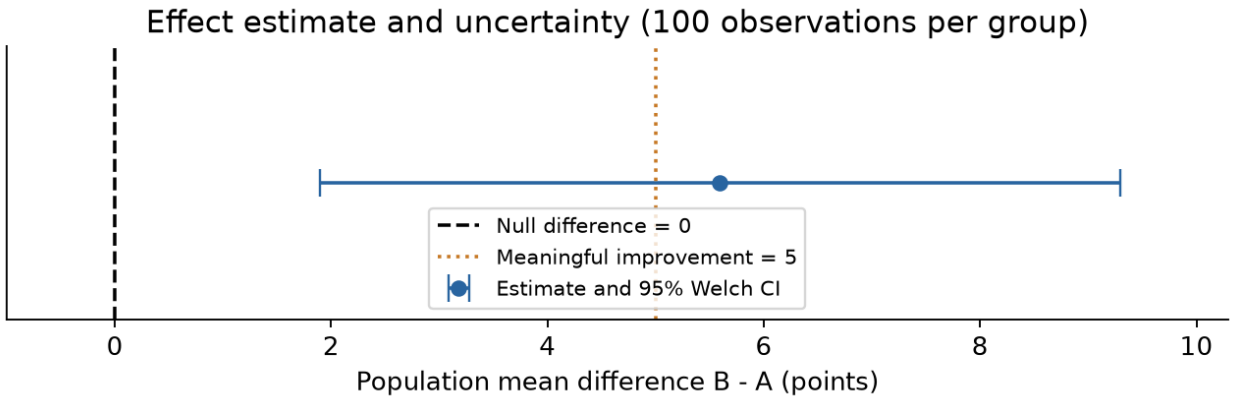
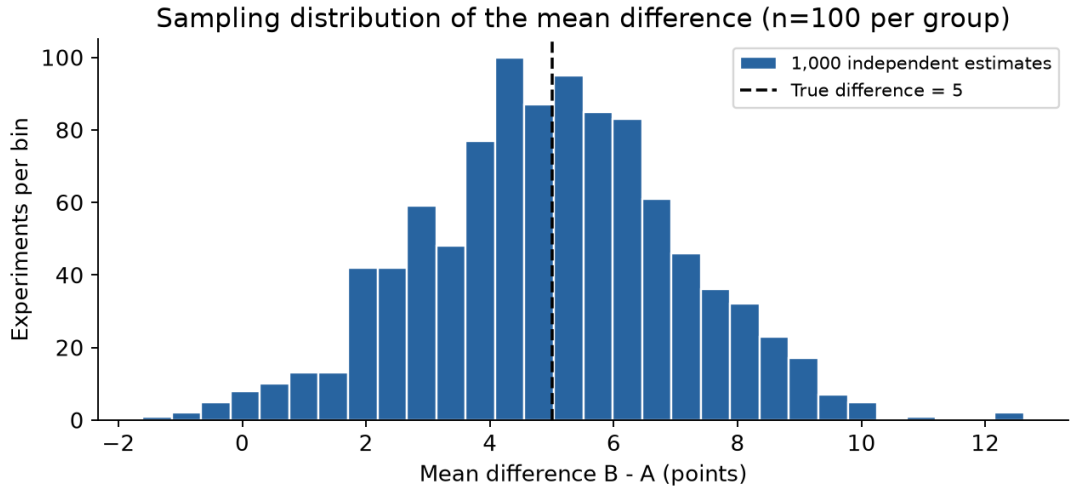


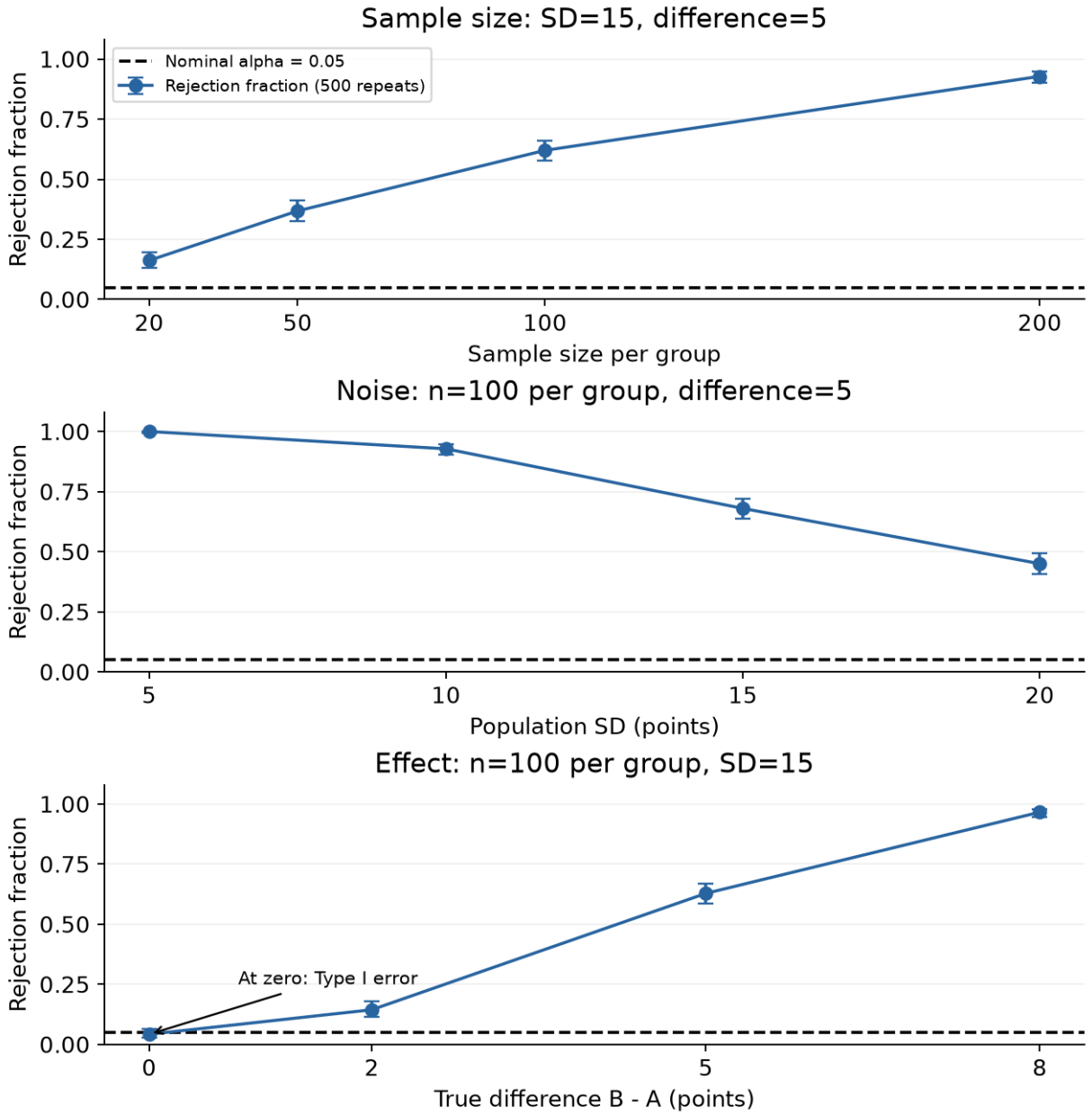
Figure 2. Point estimate and 95% Welch interval. Under its assumptions, the confidence procedure covers the fixed true difference in approximately 95% of repeated samples; it is not a probability for the fixed parameter.

Sampling distribution. In 1,000 independent experiments at n=100 per group, the mean estimate is 4.979 and empirical SD is 2.082. The theoretical SE=sqrt(15 squared/100 + 15 squared/100)=2.121. At n=200 per group it is 1.500. Larger samples reduce estimation noise.



Power: sample size, noise and effect

Each point uses 500 independent Welch tests at alpha=0.05. Change one factor at a time from n=100 per group, SD=15, difference=5. Vertical bars show 95% Wilson binomial intervals for the rejection probability. MC SE=sqrt(p-hat x (1-p-hat)/500). Even 500/500 rejections do not establish that true power equals 100%.

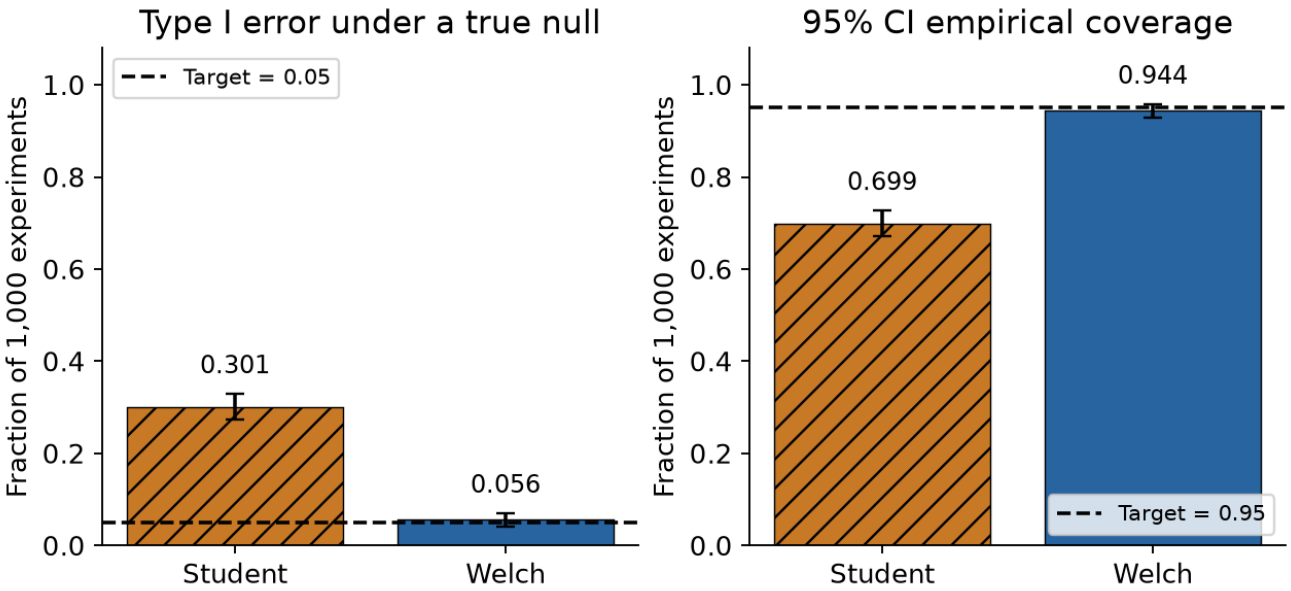


Executed rates. n=20/50/100/200: 16.2% / 36.8% / 62.0% / 92.8%.
SD=5/10/15/20: 100.0% / 92.8% / 68.0% / 45.0%.
Difference=0/2/5/8: 4.2% / 14.4% / 62.8% / 96.6%.

More observations, lower noise and larger effects increase detection in these conditions. At difference=0 the rejection rate is Type I error, not power to detect a true effect. Repeated baseline conditions use independent draws, so their rates can differ.

Counterexample, checks and limitations

Violation. Equal population means (70), independent normal draws: A has n=25, SD=20; B has n=100, SD=5. This violates Student's equal-variance assumption. Both methods use the same samples in each of 1,000 repetitions. Student uses the pooled variance and df=123.



Results. Student: Type I error=30.1%, 95% CI coverage=69.9%. Welch: Type I error=5.6%, coverage=94.4%. Bars show approximately +/-1.96 Monte Carlo SE. Coverage and non-rejection are complementary for these matching intervals and tests.

Why the answer changes. The large low-variance group dominates the pooled variance, but the small high-variance group dominates uncertainty in the difference. The true SE is 4.031; substituting population variances into the pooled formula gives only 2.216. Student can underestimate uncertainty and inflate false positives. Welch separates the two variances and sample sizes and substantially reduces this problem here.

Checks. Manual formulas match SciPy, including every counterexample CI. CSV round trips and repetition counts pass. The zero-effect rate is 4.2%; its exact binomial 95% interval is [0.026, 0.063]. Clean-kernel execution and reproducibility checks are recorded in VERIFICATION_LOG.md.

Limits. Finite repetition counts introduce Monte Carlo variation. Welch is approximate and does not correct dependence, selection bias or all outlier problems. Normal scores are unbounded; real scores may be bounded or skewed. Only the stated conditions were studied. Statistical significance alone does not prove practical importance.

Reference and review. The student must personally review the methods, results and course submission requirements. Method reference: SciPy documentation, scipy.stats.ttest_ind (https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_ind.html).